

ITSAR 2

Endterm Lab1 Activity Documentation

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Submitted To:

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```
PS D:\rovin\Projects\2025-2026\ITSAR2\etl-lab> node etl-challenge.js
--- STARTING ETL PIPELINE ---
[EXTRACT] Fetching: https://jsonplaceholder.typicode.com/users
[EXTRACT] Got 10 records.
[EXTRACT] Fetching: https://jsonplaceholder.typicode.com/posts
[EXTRACT] Got 100 records.
[TRANSFORM] Processing and matching user posts...
[TRANSFORM] Clean records matching Southern Hemisphere (lat < 0): 7



| (index) | id | name                    | username           | email                      | phone            | city            | zipcode      | lat      | lng       | company              | postCount |
|---------|----|-------------------------|--------------------|----------------------------|------------------|-----------------|--------------|----------|-----------|----------------------|-----------|
| 0       | 1  | 'Leanne Graham'         | 'bret'             | 'sincere@april.biz'        | '1-770-736-8031' | 'Gwenborough'   | '92998-3874' | -37.3159 | 81.1496   | 'Romaguera-Crona'    | 10        |
| 1       | 2  | 'Ervin Howell'          | 'antonette'        | 'shanna@melissa.tv'        | '010-692-6593'   | 'Wisokyburgh'   | '90566-7771' | -43.9589 | -34.4618  | 'Deckow-Crist'       | 10        |
| 2       | 3  | 'Clementine Bauch'      | 'samantha'         | 'nathan@yesenia.net'       | '1-463-123-4447' | 'McKenziehaven' | '50598-4157' | -68.6102 | -47.0653  | 'Romaguera-Jacobson' | 10        |
| 3       | 5  | 'Chelsey Dietrich'      | 'kamren'           | 'lucio.hettinger@annie.ca' | '(254)954-1289'  | 'Roscoeview'    | '33263'      | -31.8129 | 62.5342   | 'Keebler LLC'        | 10        |
| 4       | 6  | 'Mrs. Dennis Schulist'  | 'leopoldo_corkery' | 'karley_dach@jasper.info'  | '1-477-935-8478' | 'South Christy' | '23505-1337' | -71.4197 | 71.7478   | 'Considine-Lockman'  | 10        |
| 5       | 8  | 'Nicholas Runolfsson V' | 'maxine_nienow'    | 'shervood@rosamond.me'     | '586.493.6943'   | 'Allayview'     | '45169'      | -14.399  | -120.7677 | 'Abernathy Group'    | 10        |
| 6       | 10 | 'Clementina DuBuque'    | 'mariah.stanton'   | 'rey.padberg@karina.biz'   | '024-648-3084'   | 'Lebsackbury'   | '31428-2261' | -38.2386 | 57.2232   | 'Hoeger LLC'         | 10        |



[LOAD] Writing 7 rows to CSV...
[LOAD] Saved to output.csv successfully.
--- ETL PIPELINE COMPLETE ---
PS D:\rovin\Projects\2025-2026\ITSAR2\etl-lab>
```

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Paste Cut Copy Format Painter Clipboard Font Alignment Number Styles

Calibri 11 A⁺ B I U Wrap Text Merge & Center General \$ % Conditional Formatting Format as Table Normal Bad Check Cell Explanatory ...

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

A	B	C	D	E	F	G	H	I	J	K	L
id	name	username	email	phone	city	zipcode	lat	lng	company		
1	Leanne Graham	bret	sincere@april.biz	1-770-736-8031	Gwenborough	92998-3874	-37.3159	81.1496	Romaguera-Crona		
2	Ervin Howell	antonette	shanna@melissa.tv	010-692-6593	Wisokyburgh	90566-7771	-43.9509	-34.4618	Deckow-Crist		
3	Clementine Bauch	samantha	nathan@yesenia.net	1-463-123-4447	McKenziehaven	59590-4157	-68.6102	-47.0653	Romaguera-Jacobson		
4	Patricia Lebsack	karianne	julianne.oconner@kory.org	493-170-9623	South Elvis	53919-4257	29.4572	-164.299	Robel-Corkery		
5	Chelsey Dietrich	kamren	lucio_hettinger@annie.ca	(254)954-1289	Roscoeview	33263	-31.8129	62.5342	Keebler LLC		
6	Mrs. Dennis Schulist	leopoldo_corkery	karley_dach@jasper.info	1-477-935-8478	South Christy	23505-1337	-71.4197	71.7478	Considine-Lockman		
7	Kurtis Weissnat	elwyn.skiles	telly.hoeger@billy.biz	210-607-6132	Howemouth	58804-1099	24.8918	21.8984	Johns Group		
8	Nicholas Runolfsson V	maxime_nienow	sherwood@rosamond.me	586.493.6943	Aliyaview	45169	-14.399	-120.768	Abernathy Group		
9	Glenna Reichert	delphine	chaim_mcdermott@dana.io	(775)976-6794	Bartholomebury	76495-3109	24.6463	-168.889	Yost and Sons		
10	Clementina DuBuque	moriah.stanton	rey.padberg@karina.biz	024-648-3804	Lebsackbury	31428-2261	-38.2386	57.2232	Hoeger LLC		

Task 1

Q1. What fields are nested objects (not plain values)?

Answer: The nested fields are address (which also has a smaller geo object hidden inside it) and company. They are objects containing multiple pieces of information rather than a single piece of text or number.

Q2. What is the value of `data[0]. address.city`?

Answer: Gwenborough

Task 2

Q1. What does `.split(' ')[0]` do to the phone number?

Answer: It cuts the phone number at the first blank space and keeps only the front part. This removes extra messy text like extensions (for example, chopping off x56442) so you are left with just the main phone number.

Q2. Why do we use `parseFloat()` on lat and lng?

Answer: The website sends the latitude (lat) and longitude (lng) as text (strings). We use `parseFloat()` to convert that text into actual math numbers with decimals so computers can read them correctly as map coordinates.

Q3. The filter step removes records without '@' in the email. How many records passed the filter?

Answer: 10 records passed the filter. All the users from the website have a valid email with an @ symbol, so nobody was thrown away.

Task 3

Q1. How many columns does the CSV have? List them.

Answer: It has 10 columns. They are:

- id, name, username, email, phone, city, zipcode, lat, lng, and company.

Q2. What is the company name of the user with id = 5?

Answer: Keebler LLC. (Note: This is the real data returned by the online website for user number 5).

Q3. What does `fs.writeFileSync()` do? What would happen if you used `fs.appendFileSync()` instead?

Answer: * fs.writeFileSync() creates a brand-new file or completely overwrites the old one if it already exists. This ensures your spreadsheet is fresh and clean every time you run the script.

- If you used fs.appendFileSync(), it would add the new data to the very bottom of the old file without erasing it. Every time you ran the script, the file would get longer and messier, stacking duplicate rows and headers on top of each other.

Challenge Exercise

Code:

```
// etl.js - Upgraded Challenge Version
const fs = require('fs');
const path = require('path');

const USERS_API_URL = 'https://jsonplaceholder.typicode.com/users';
const POSTS_API_URL = 'https://jsonplaceholder.typicode.com/posts';

// =====
// 1. EXTRACT (Challenge B: Keeps errors throwing so main handles them)
// =====
async function extract(url) {
  console.log(`[EXTRACT] Fetching: ${url}`);
  const response = await fetch(url);
  if (!response.ok) {
    throw new Error(`HTTP error! Status: ${response.status} on URL: ${url}`);
  }
  const data = await response.json();
  console.log(`[EXTRACT] Got ${data.length} records.`);
  return data;
}

// =====
// 2. TRANSFORM (Includes Challenge A & Challenge C)
// =====
function transform(users, posts) {
  console.log(`[TRANSFORM] Processing and matching user posts...`);

  const rows = users.map(user => {
    // CHALLENGE A: Count how many posts belong to this user's ID
    const userPosts = posts.filter(post => post.userId === user.id);
    const postCount = userPosts.length;

    return {
      id: user.id,
      name: user.name,
      username: user.username.toLowerCase(),
      email: user.email.toLowerCase(),
      phone: user.phone.split(' ')[0],
      city: user.address.city,
      zipcode: user.address.zipcode,
      lat: parseFloat(user.address.geo.lat),
      lng: parseFloat(user.address.geo.lng),
      company: user.company.name,
      postCount: postCount // CHALLENGE A: Added column
    };
  });

  // CHALLENGE C: Filter to only keep users with a negative latitude (Southern Hemisphere)
  const southernHemisphereUsers = rows.filter(r => r.lat < 0);

  console.log(`[TRANSFORM] Clean records matching Southern Hemisphere (lat < 0):`,
    southernHemisphereUsers.length);
  return southernHemisphereUsers;
}

// =====
// 3. LOAD
// =====
function load(rows) {
  console.log(`[LOAD] Writing ${rows.length} rows to CSV...`);

  if (rows.length === 0) {
    console.log(`[LOAD] No records to save.`);
    return;
  }

  const headers = Object.keys(rows[0]);
  const lines = rows.map(row => headers.map(h => `${row[h]}`).join(','));
  const csv = [headers.join(','), ...lines].join('\n');

  const filePath = path.join(__dirname, 'output.csv');
  fs.writeFileSync(filePath, csv, 'utf8');
  console.log(`[LOAD] Saved to output.csv successfully.`);
}

// =====
// MAIN CONTROL (Challenge D: Error Handling Wrap)
// =====
async function main() {
  try {
    console.log(`--- STARTING ETL PIPELINE ---`);

    // Challenge A: Extracting from both endpoints
    const rawUsers = await extract(USERS_API_URL);
    const rawPosts = await extract(POSTS_API_URL);

    // Pass both sets of data into transform
    const cleanData = transform(rawUsers, rawPosts);

    // Preview the outcome
    console.table(cleanData);

    // Load into CSV
    load(cleanData);

    console.log(`--- ETL PIPELINE COMPLETE ---`);
  } catch (error) {
    // CHALLENGE B: Friendly error catch message
    console.error(`\n[ETL PIPELINE CRASHED!]\n`);
    console.error(`Reason: ${error.message}`);
    console.error(`Please verify your internet connection or check if your API URL endpoints are spelled correctly.\n`);
  }
}

main();
```

```

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| 2       | 3  | 'Clementine Bauch'      | 'samantha'       | 'nathan@yesenia.net'       | '1-463-123-4447' | 'McKenziehaven' | '59590-4157' | -68.0102 | -47.0653  | 'Romaguera-Jacobson' | 10        |
| 3       | 5  | 'Chelsey Dietrich'      | 'karen'          | 'lucio.hettinger@annie.ca' | '(254)954-1289'  | 'Roscoeview'    | '33663'      | -31.8129 | 62.5342   | 'Keebler LLC'        | 10        |
| 4       | 5  | 'Doris Anderson'        | 'skuhlit'        | 'kavita.dach@jasper.info'  | '1-477-935-8478' | 'South Christy' | '23505-1337' | -71.4037 | 71.7473   | 'Considine-Lockman'  | 10        |
| 5       | 8  | 'Nicholas Runolfsson V' | 'maxime_nienow'  | 'shermoo@rosamond.me'      | '586.493.6943'   | 'Aliyaview'     | '45169'      | -14.399  | -120.7677 | 'Abernathy Group'    | 10        |
| 6       | 10 | 'Clementina DuBuque'    | 'moriah.stanton' | 'rey.padberg@karina.biz'   | '024-648-3884'   | 'Lebsackbury'   | '31428-2261' | -38.2386 | 57.2232   | 'Hoeger LLC'         | 10        |



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| 1       | 2  | 'Ervin Howell'          | 'antonette'        | 'shanna@melissa.tv'        | '1-877-659-5959' | 'Wisokyburgh'   | '90566-7771' | -43.9589 | -34.4618  | 'Deckow-Crist'       | 10        |
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| 5       | 8  | 'Nicholas Runolfsson V' | 'maximo.niemov'    | 'sherwood@rosamond.me'     | '586.493.6943'   | 'Aliyaview'     | '45169'      | -14.399  | -120.7677 | 'Abernathy Group'    | 10        |
| 6       | 10 | 'Clementina DuBuque'    | 'coriah.stanton'   | 'rey.padberg@karina.biz'   | '624-648-3804'   | 'Lebsackbury'   | '31428-2261' | -38.2386 | 57.2232   | 'Hoeger LLC'         | 10        |



[LOAD] Writing 7 rows to CSV...

[!! PIPELINE CRASHED !!]
Reason: EBUSY: resource busy or locked, open 'D:\rovin\Projects\2025-2026\ITSAR2\etl-lab\output.csv'
Please verify your internet connection or check if your API URL endpoints are spelled correctly.
```

[illegible]